

Tutorial 7(a)

Newton's Law of Gravitation & Gyroscope

1. Collision of two satellites

A 400 kg satellite in a 1000 km altitude circular orbit is going to collide with a smaller 100 kg satellite travelling in the same orbit but in the opposite direction. After the collision, they stick together into a single piece of space debris. Determine whether the space debris will continue to orbit around earth or crash into the earth. Use radius of the earth as 6400 km and mass of earth as 6×10^{24} kg.

2. Earth Tunnel to its Centre

Assume Earth is a uniform sphere. Consider a tunnel being dug from the surface to the centre of the Earth. A mass of 7 kg is dropped from rest into the tunnel. What is the speed of the mass when it reaches the centre of the Earth? Take the radius of the Earth to be 6370 km. Assuming the mass do not touch the sides of the tunnel while falling and ignore air resistance.

3. Space Elevator 天梯

It is proposed to make a space elevator by placing a long uniform rod extending from just above the surface of the earth (not fixed to the surface) out to radius nR_E (R_E is the radius of the earth). Prove that the rod can remain above the same point on the equator all the time in a circular orbit if

$$n^2 + n = \frac{8\pi G\rho}{3\omega_0^2}$$

where ρ is the density of the earth and ω_0 is the angular speed of the Earth.

4. Orbiting round an alien planet

A 800 kg satellite orbits circularly around a uniform spherical alien planet at a radius of 8000 km with a period of 6200 sec. Find the total energy of the satellite if it orbits around the alien planet circularly at a radius of 10500 km.

Answer: -2×10^{10} J (approximate)

5. Aphelion and Perihelion 远日点与近日点

A comet is in an elliptical orbit about the Sun has a velocity of 10 km/sec at aphelion (furthest distance from the Sun) and 80 km/sec at perihelion (nearest distance from the Sun). The mass of the Sun is 2×10^{30} kg. Find the aphelion distance for the comet.

Answer: 2.96×10^{11} m

6. Minimum Period

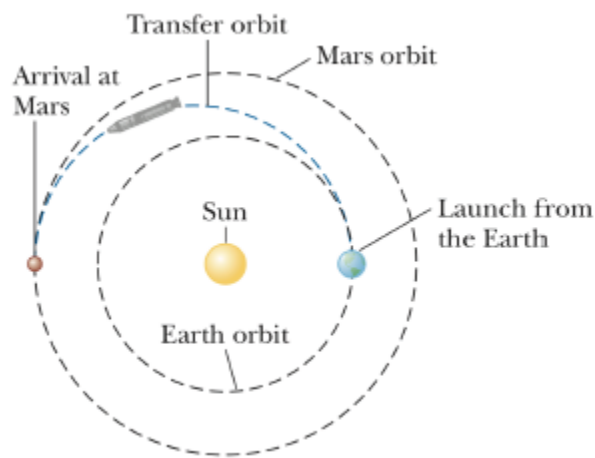
Show that the minimum period for a satellite in orbit around a spherical planet of uniform density ρ is

$$T_{min} = \sqrt{\frac{3\pi}{G\rho}}$$

independent of any radius.

7. Changing from a Circular to an Elliptical orbit

A minimum energy transfer orbit to an outer planet consists of putting a spacecraft on an elliptical trajectory with the departure planet corresponding to the perihelion of the ellipse, or the closest point to the Sun and the arrival planet at the aphelion, or the farthest point from the Sun. Consider the case of Earth to Mars transfer as shown in the figure. Treat both planets as travelling in circular orbits and the Earth's orbital radius is 1.50×10^8 km and the Mars orbital radius is 2.30×10^8 km and the Sun's mass to be 2.00×10^{30} kg. Calculate how long the spacecraft will take to arrive at Mars, giving your answer in years.



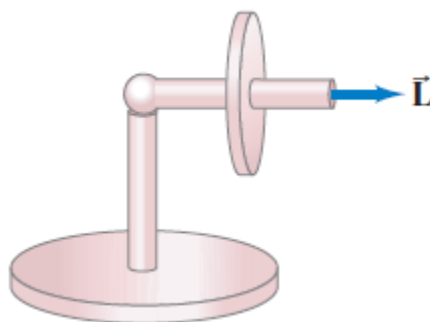
8. Maximum Gravitational Force between two broken masses

A mass M is broken into two parts with one part having mass xM and the other part having mass $(1 - x)M$ where $0 < x < 1$. The two parts are separated by a certain distance. For a given separation distance, determine the value of x for which the gravitational attraction between the two pieces is a maximum. You can treat each part as a point mass.

Answer: 1/2

9. Gyroscope with an added point mass

Suppose the solid wheel as shown in the figure below has a mass of 300 g and rotates at 85 rad/s; it has radius 6 cm and is mounted at the center of a horizontal thin axle 25 cm long. At what rate does the axle precess? If a mass equal to half the mass of the wheel is placed at the free end of the axle, what will be the precession rate now? Treat the extra mass as insignificant in size.



Answer: 8.0 rad/s, 16 rad/s

10. Grain Mill 谷物磨坊

In a grain mill, grain is grounded by a massive wheel that rolls without slipping in circle on a flat horizontal millstone driven by a vertical shaft. The rolling wheel has mass M , radius b and is constrained to roll in a horizontal circle of radius R at angular speed Ω as shown in the figure below. The wheel pushes down on the lower millstone with a force equal to twice its weight (normal force). The mass of the axle of the wheel can be neglected. What is the precessional angular frequency Ω ?

